

Cybersecurity

ML and AI in Cybersecurity Offense

Lesson 2.3



Learning Objectives

By the end of today's class, you will:

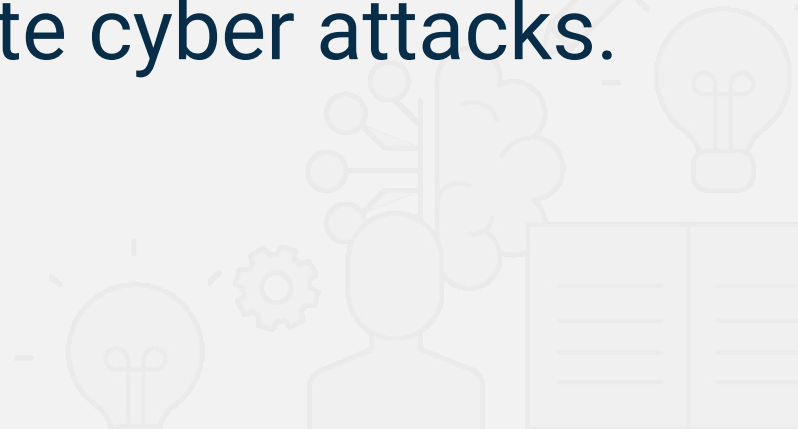
- 1 Understand how AI is used in cybersecurity offense.
- 2 Learn about specific AI and ML techniques for offensive security.
- 3 Gain hands-on experience with AI tools used in cybersecurity offense.



AI and ML Techniques for **Cybersecurity** **Offense**



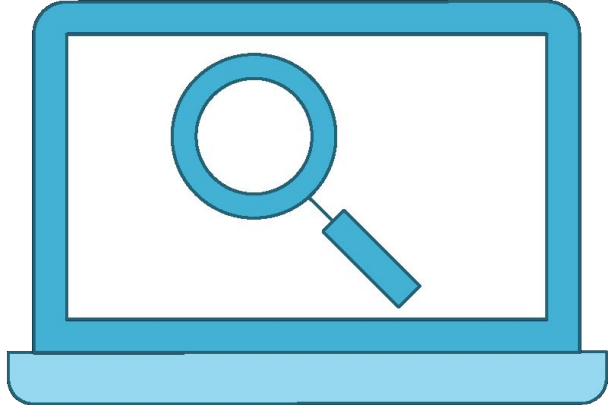
Offensive security involves using AI and ML techniques to simulate cyber attacks.



Common Applications Using AI in Offensive Cybersecurity

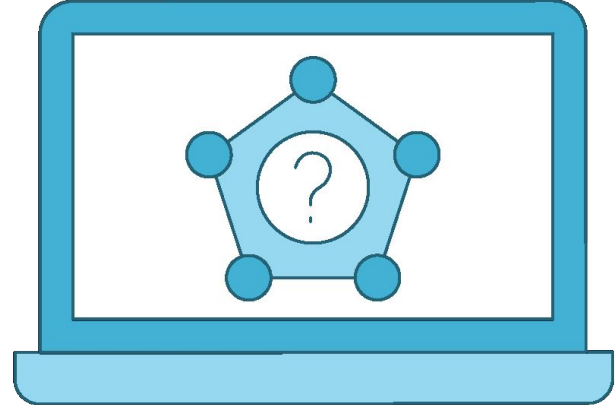
Penetration Testing

Identifying specific vulnerabilities in a system or application.



Red Teaming

Testing overall security posture and response capabilities.

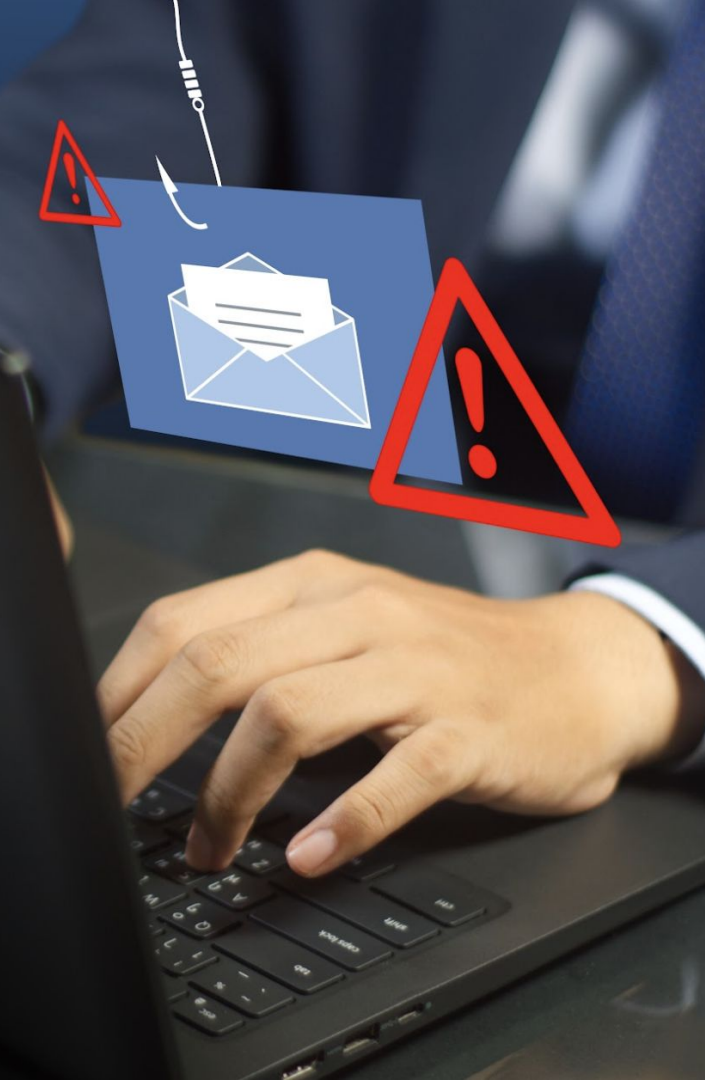




Automated Exploitation Generation

Helps identify vulnerabilities

- Analyzes software
- Generates exploits
- Speeds up process



Phishing Campaign Automation

Increases effectiveness of phishing campaigns

- Crafts and sends phishing emails
- Adjusts based on recipient response
- Adapts and improves overtime



Social Engineering

Crafts personalized attack vectors

- Gathers and analyzes detailed information
- Uses various social media platforms
- Designs effective attacks

Examples of AI and ML Applications in Offensive Security

Penetration testing tools

AI-driven tools like automated vulnerability scanners identify and exploit system weaknesses, providing a comprehensive security assessment.

Phishing simulation platforms

AI-powered platforms simulate phishing attacks to train employees on recognizing and responding to phishing attempts.

Advanced persistent threat (APT) simulations

AI tools that simulate APTs test organizational defenses against prolonged and sophisticated cyber attacks.



AI and ML for Automated Vulnerability Scanning



Automated vulnerability scanning involves using specialized tools to identify security vulnerabilities in systems, networks, and applications.

Understanding Automated Vulnerability Scanning

01

Unpatched software



02

Misconfigurations



03

Potential exploit points



AI and ML Enhances Vulnerability Scanning

-  AI can assess the potential impact and exploitability of vulnerabilities, and prioritize remediation efforts accordingly.
-  AI and ML improve the effectiveness of vulnerability scans.
-  Prioritization of vulnerabilities helps resource optimization for remediation.

Key AI and ML Techniques for Vulnerability Scanning

Pattern recognition

AI can identify known vulnerabilities by matching patterns in code against a database of known issues.

Anomaly detection

AI can identify unusual patterns that could indicate novel vulnerabilities.

Predictive analysis

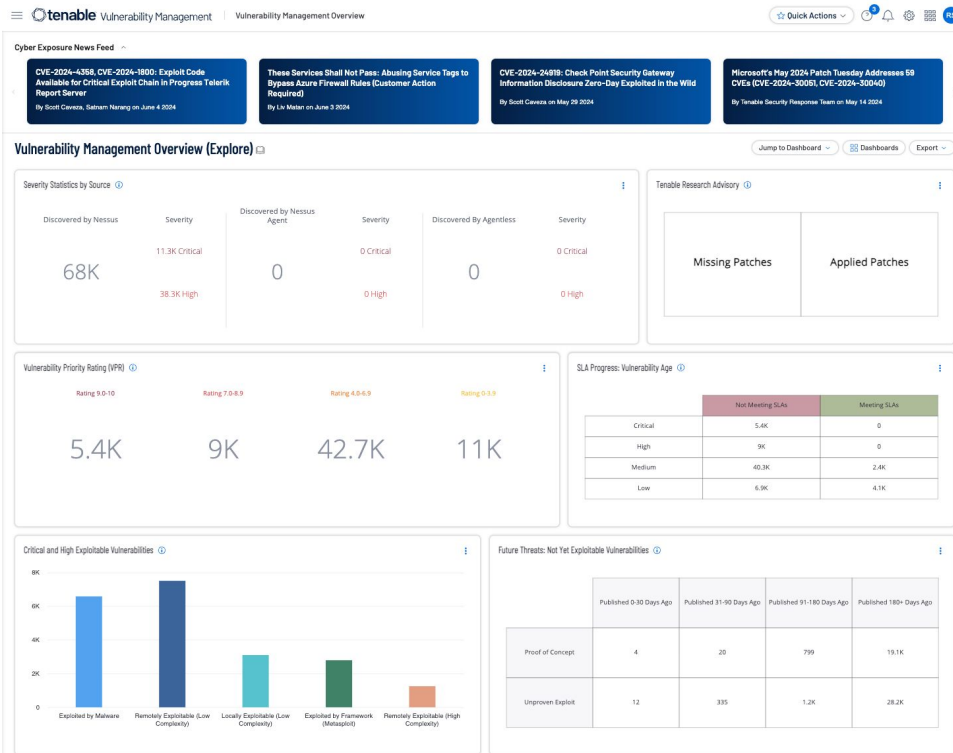
Historical data is used to forecast future potential vulnerabilities and prioritize them based on likelihood of being exploited.



AI and ML Applications in Vulnerability Scanning

Tenable.io & Rapid7 InsightVM

- Identification of weaknesses
- Prioritization of vulnerabilities
- Continuous monitoring
- Automated assessments

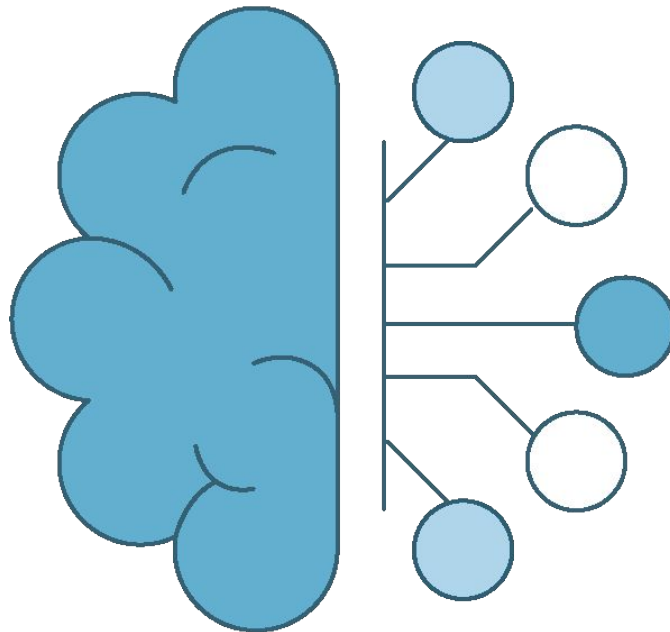


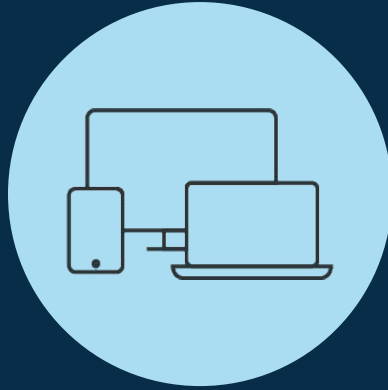


AI and ML Applications in Vulnerability Scanning

Automated Scanners with AI Integration

- Assessment of security posture
- Real-time assessments
- Adaptation to new threats
- Up-to-date information





Instructor Demonstration

APT Threat Analysis



ChatGPT: “Summarize APT 29.”



APT 29, also known as "Cozy Bear," is a sophisticated cyber espionage group believed to be associated with the Russian government. This group has been active for several years and is known for its targeted attacks primarily against government, diplomatic, think-tank, healthcare, and energy organizations to gather intelligence. Some key characteristics of APT 29 include:

1. **Sophistication:** APT 29 uses a variety of advanced techniques, including spear-phishing, malware, and encrypted communication channels to avoid detection.
2. **Long-term Focus:** Unlike many groups that launch quick-hit attacks, APT 29 prefers long-term operations, maintaining a quiet presence in the infected networks to continuously gather data over extended periods.
3. **Adaptability:** The group frequently updates its TTPs (Tactics, Techniques, and Procedures) to include the latest exploits and to adapt to cybersecurity measures, showing a high degree of adaptability and learning.
4. **High-Profile Attacks:** APT 29 has been implicated in several high-profile cyber operations, including alleged interference in the 2016 US presidential election and various attacks on COVID-19 vaccine research facilities.

The focus on stealth and the use of sophisticated techniques make APT 29 a significant threat in the realm of cybersecurity, especially for entities holding sensitive political, diplomatic, or strategic intelligence.



Activity:

Analyzing APTs with AI

In this activity, you will be researching APT threats targeting financial organizations and summarizing the results using ChatGPT.

Suggested Time:
20 Minutes





Time's up!
Let's review

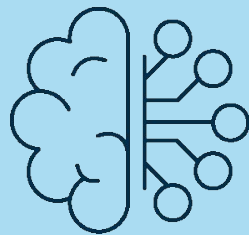


Questions?





Break
15 mins



AI Techniques for Automated Exploit Generation



Exploit generation involves creating code or techniques designed to take advantage of vulnerabilities in software, systems, or networks.



The primary goal of **exploit generation** is to demonstrate the potential impact of a vulnerability and to help security professionals understand how it can be mitigated.



How does AI and ML
enhance **exploit**
generation?



A white robotic hand is positioned over a black computer keyboard. In the background, a computer monitor displays a code editor with syntax-highlighted code in various colors (green, blue, red) on a dark background. The scene is dimly lit, with the primary light source coming from the monitor.

AI and ML Enhances Exploit Generation

Key benefits:

- Speed and efficiency
- Accuracy
- Thorough analysis
- Deeper insights

Complex vulnerabilities: AI can identify complex and subtle vulnerabilities that human analysts might miss.

Key AI and ML Techniques for Exploit Generation

Machine learning for vulnerability detection

Using ML to identify vulnerabilities

Example: Training ML on a dataset of known vulnerabilities

Automated exploit development

Using AI to generate exploits for vulnerabilities

Example: Using reinforcement learning to optimize exploit code

Fuzzing

Testing technique inputting random or unexpected data

Example: Using genetic algorithms to evolve inputs to test software issues

Examples of AI/ML Applications in Exploit Generation

Advanced AI modules can help in exploit generation

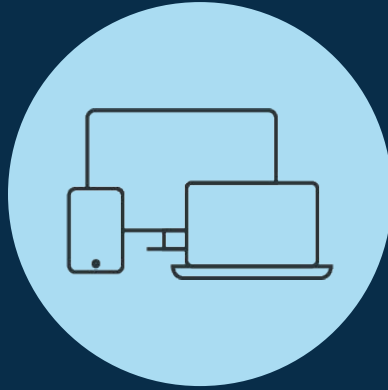
OpenAI's GPT-4:

- Analyzes vulnerabilities
- Generates exploit code

AI-powered fuzzing tools:

- Tests for weaknesses
- Generates reports
- Provides actionable insights





Instructor Demonstration

AI and ML Tools for Exploit Generation

Common Vulnerabilities and Exposure

cve.mitre.org



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TOTAL CVE Records: **237725**

NOTICE: Transition to the all-new CVE website at WWW.CVE.ORG and [CVE Record Format JSON](#) are underway.

NOTICE: Support for the legacy CVE download formats will end on June 30, 2024.
New CVE List download format is available now.

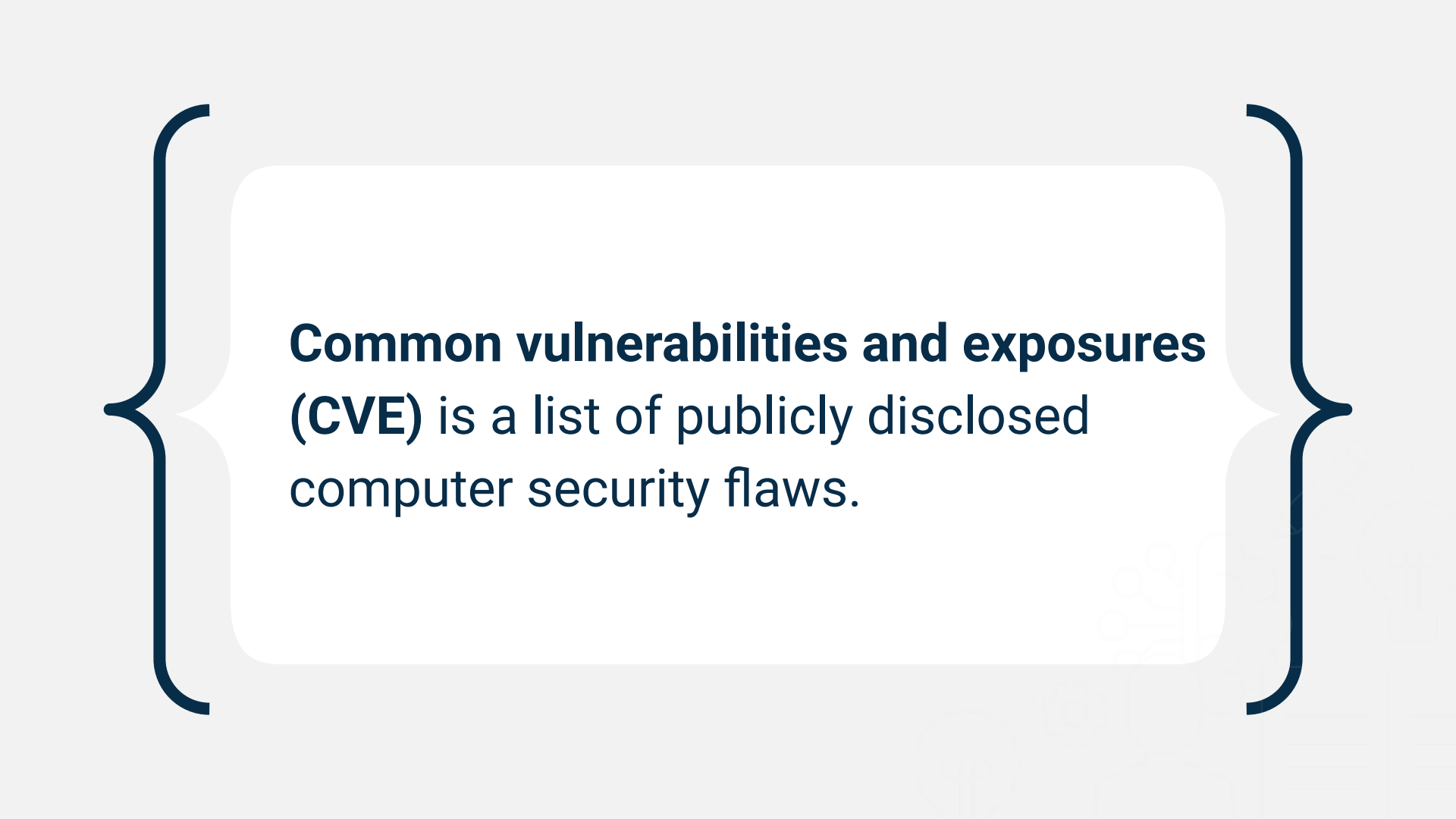
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Attention: CVE Records now include product versions & more on the new CVE.ORG website. Learn about the new and improved [CVE Record Format](#).

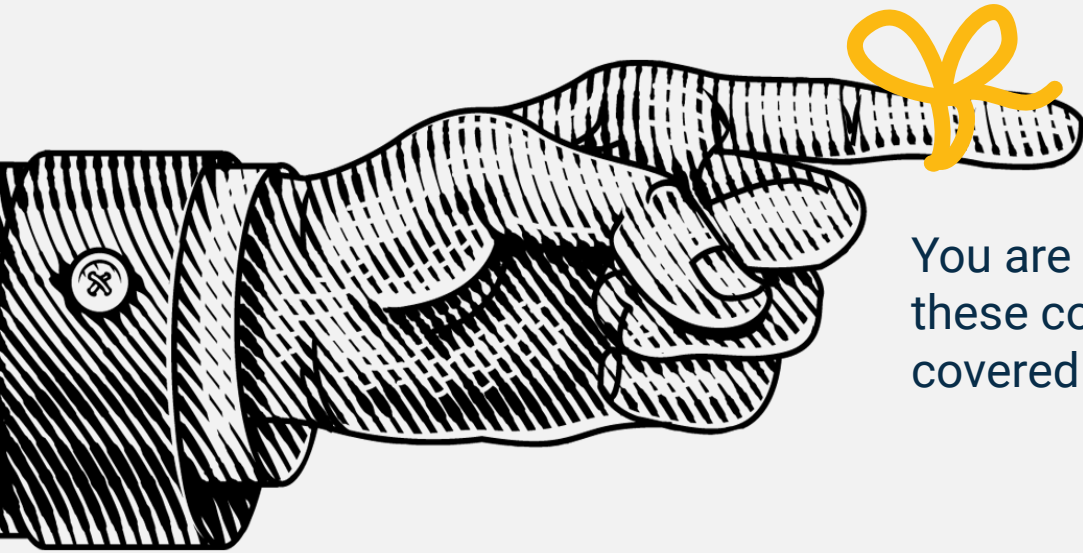
Submit



Common vulnerabilities and exposures (CVE) is a list of publicly disclosed computer security flaws.

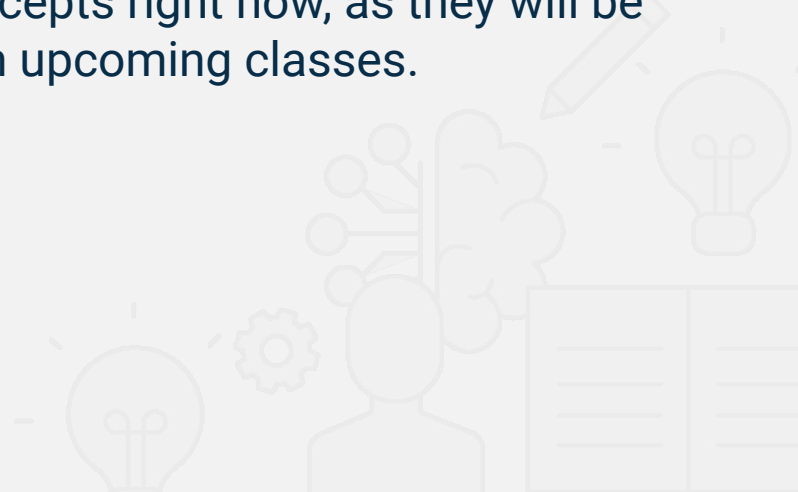
Common Vulnerabilities and Exposures

-  Provides a standardized identifier for a vulnerability
-  Widely used in the cybersecurity industry to quickly and accurately share information about vulnerabilities
-  Information relates to nature of vulnerabilities, systems affected, and degree of criticality
-  Tracks vulnerability metrics and manages patches and updates to mitigate threats



Remember,

You are not expected to understand all of these concepts right now, as they will be covered in upcoming classes.





Activity:

Generating Exploits with AI

In this activity, you will be using ChatGPT and WhiteRabbitNeo to generate an exploit to analyze.

Suggested Time:
20 Minutes





Time's up!
Let's review



Questions?





Let's recap



Recap

ML and AI in Cybersecurity Offense

01

We discussed how AI is used in cybersecurity offense.

02

You learned about specific AI/ML techniques for offensive security.

03

You explored AI tools used in cybersecurity offense.



Next class

We will discuss basic **Linux systems** and the command line, looking into distros and file structures.



Thank you